UNIVERSIDAD DE LA SIERRA JUÁREZ

CARRERA:
LICENCIATURA EN INFORMÁTICA

ASIGNATURA:
Investigación de Operaciones

NOMBRE DEL TEMA:
Respuestas del examen ordinario

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Alumno:
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Semestre: Octavo

Problema 1

$$\rho = \frac{\lambda}{s\mu} = \frac{5}{3 \cdot 8} = \frac{5}{24}$$

a) $$P_0 = \left(\frac{s^0 \rho^s}{s!(1-\rho)} + \sum_{n=0}^{s-1} \frac{(s\rho)^n}{n!} \right)^{-1} = \left(\frac{3^3 \rho^3}{3!(1-\rho)} + \sum_{n=0}^{2} \frac{(3\rho)^n}{n!} \right)^{-1}$$

$$\left(\frac{3^3 \rho^3}{3!(1-\rho)} + \frac{(3\rho)^0}{0!} + \frac{(3\rho)^1}{1!} + \frac{(3\rho)^2}{2!} \right)^{-1}$$

$$= \left(\frac{125}{2432} + 1 + \frac{5}{8} + \frac{25}{128} \right)^{-1} = \frac{304}{569} = 0.53428$$

b) $$L_q = \frac{s^s \rho^{s+1} P_0}{s!(1-\rho)^2} = \frac{3^3 \rho^4 \frac{304}{569}}{3!(1-\rho)^2} = \frac{302}{41791} \approx 0.00722643 \text{ Cliente}$$

c) $$W_q = \frac{L_q}{\lambda} = \frac{302}{5 \cdot 41791} = \frac{52}{35979} \approx 0.00144529h$$

d) $$W = W_q + \frac{1}{\mu} = \frac{52}{35979} + \frac{1}{8} = \frac{514}{4065} \approx 0.126445h$$

e) $$L = \lambda W = 5 \cdot \frac{514}{4065} = \frac{514}{813} \approx 0.6323 \text{ Clientes}$$

Figura 1: Ejercicio 1

Problema 2

a)

Tasa llegada $\lambda = 80$ Clientes / hora

Tasa servicio $\mu = 60 / 1.2 = 5$ Clientes hora

Factor Utilizacion

$$\rho = \frac{\lambda}{5 \cdot \mu} = \frac{80}{2 \cdot 50} = 0.8$$

Probabilidad no Clientes

$$P_0 = \frac{1}{\sum_{n=0}^{1} \frac{(80/50)^n}{n!} + \frac{(80/50)^2}{2!}\left(\frac{100}{100-80}\right)} = 0,111$$

Promedio Clientes en la cola

$$L_q = \frac{(\lambda/\mu)^5 \lambda \mu}{(5-1)!\,(5\mu-\lambda)^2} P_0 = \frac{(80/50)^2 \times 80 \times 50}{1!\,(2 \times 50 - 80)^2}$$

$= 0.111 = 2,84$ Clientes

Num promedio Clientes en el sistema

$$L_s = L_q + \frac{\lambda}{\mu} = 2,84 + \frac{80}{50} = 4.44$$

Figura 2: Ejercicio 2

b) $w_a = \frac{L_a}{\lambda} = \frac{2.84}{80} = 0.035$

Tiempo estancia sistema

$w_s = w_a + \frac{1}{\mu} = 0.0355 + \frac{1}{50} = 0.0555$ horas

$= 3.33$ min

c) $P_n = \frac{(\lambda/\mu)^n}{n} P_0 = 1P_0 + \frac{1}{2} P_1 = \frac{(80/50)^1}{1!} P_0$

$= P_0 + 0.8P_0 = 1.8P_0 = 0.2$

Figura 3: Ejercicio 2

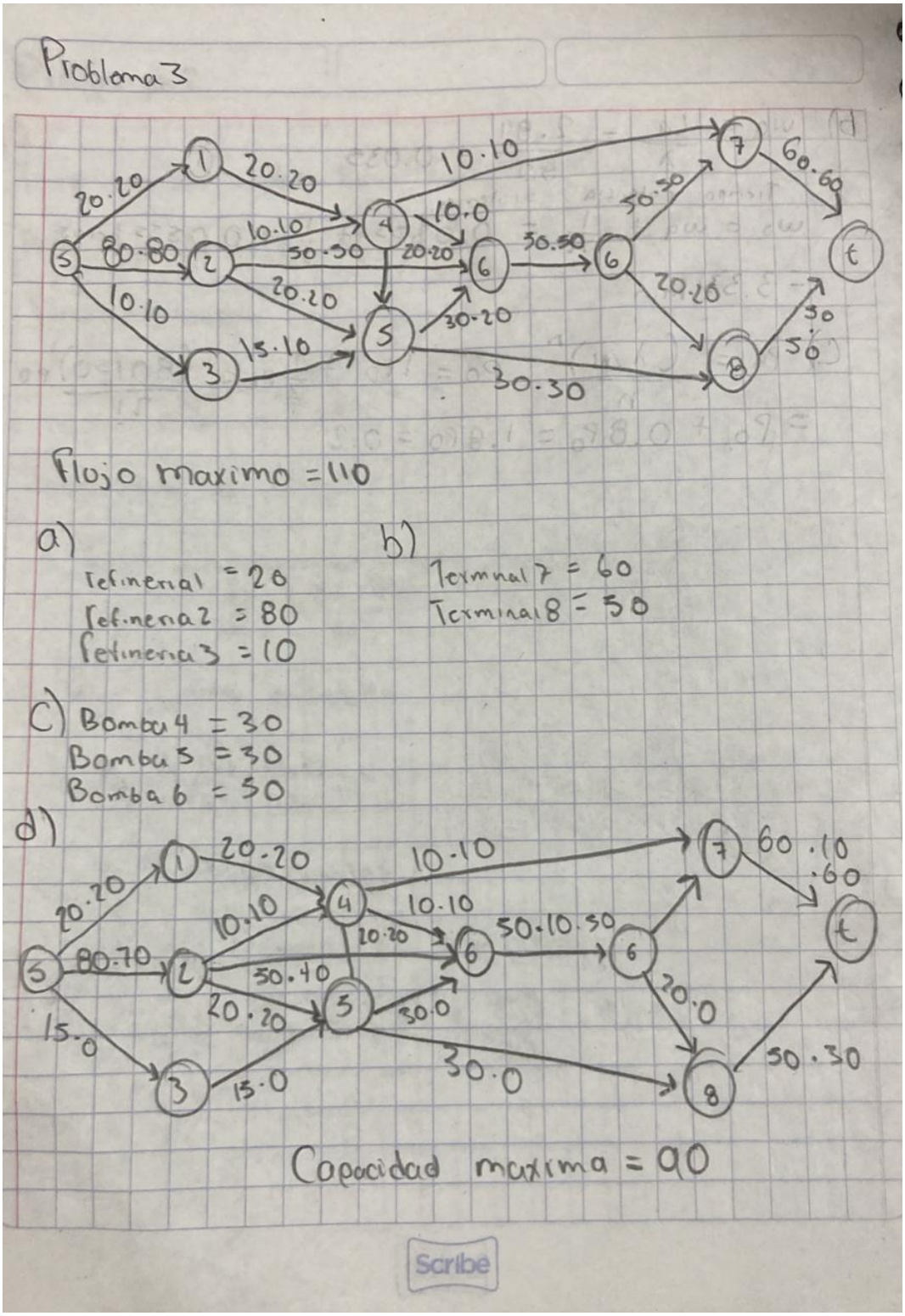


Figura 4: Ejercicio 3

Problema 5

x_1 = boldos de mano
x_2 = boldas de rasuradora
x_3 = mochilas

Funcion Objetivo

$$\text{Max. } Z = 24x + 22x_2 + 45x_3$$

Restricciones

$$2x_1 + x_2 + 3x_3 \leq 42$$
$$2x_1 + x_2 + 2x_3 \leq 40$$
$$1x + \tfrac{1}{2}x_2 + x_3 \leq 45$$

Forma estandar

$$Z - 24x_1 - 22x_2 - 45x_3 = 0$$
$$2x_1 + x_2 + 3x_3 = 42$$
$$2x_1 + x_2 + 2x_3 = 40$$
$$1x_1 + \tfrac{1}{2}x_2 + x_3 = 45$$

Solución basica

$S1 = 42$ $\quad x_1, x_2, x_3 = 0$
$S2 = 40$ $\quad Z = 0$
$S3 = 45$

Var basicas $(S1, S2, S3)$
Var no basicas (x_1, x_2, x_3)

básico	Z	x_1	x_2	x_3	S1	S2	S3	b
Z	1	-24	-22	-45	0	0	0	0
S1	0	2	1	3	1	0	0	42
S2	0	2	1	2	0	1	0	40
S3	0	1	1/2	1	0	0	1	45

Figura 5: Ejercicio 5

Basica	Z	X1	X2	X3	S1	S2	S3	b
Z	1	6	-7	0	15	0	0	630
X3	0	2/3	1/3	1	1/3	0	0	14
S2	0	2/3	1/3	0	-2/3	1	0	12
S3	0	1/3	1/6	0	-1/3	0	1	31

Basica	Z	X1	X2	X3	S1	S2	S3	b
Z	1	20	0	0	1	21	0	882
X3	0	0	0	1	1	-1	0	2
X2	0	2	1	0	-2	3	0	36
S3	0	0	0	0	0	-1/2	1	25

Respuestas

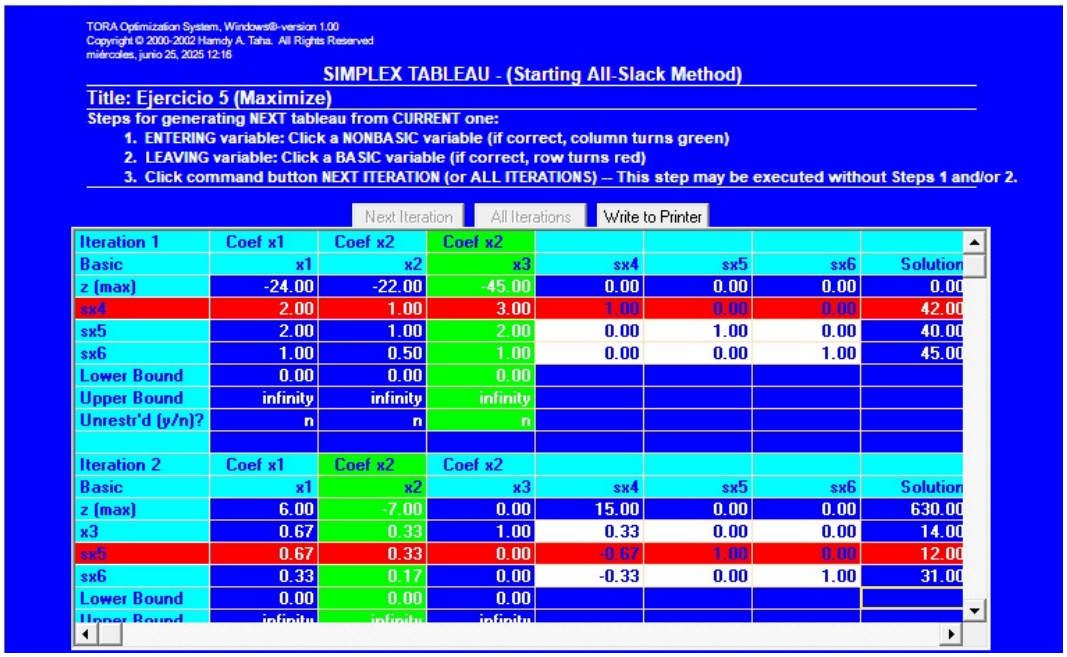
a)

X2 = 36
X3 = 2
S3 = 25
X1, S1, S2 = 0
Z = 882

b) S1 y S2 escasos

S3 Abundante

Figura 6: Ejercicio 5



TORA Optimization System, Windows®-version 1.00
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miércoles, junio 25, 2025 12:16

SIMPLEX TABLEAU - (Starting All-Slack Method)

Title: Ejercicio 5 (Maximize)

Steps for generating NEXT tableau from CURRENT one:

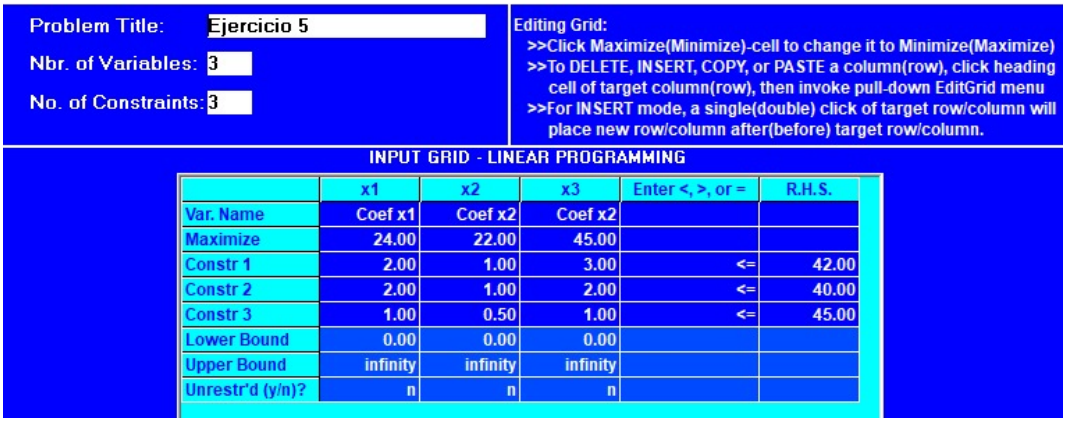
1. ENTERING variable: Click a NONBASIC variable (if correct, column turns green)
2. LEAVING variable: Click a BASIC variable (if correct, row turns red)
3. Click command button NEXT ITERATION (or ALL ITERATIONS) -- This step may be executed without Steps 1 and/or 2.

Next Iteration | All Iterations | Write to Printer

Iteration 1	Coef x1	Coef x2	Coef x2				
Basic	x1	x2	x3	sx4	sx5	sx6	Solution
z (max)	-24.00	-22.00	-45.00	0.00	0.00	0.00	0.00
sx4	2.00	1.00	3.00	1.00	0.00	0.00	42.00
sx5	2.00	1.00	2.00	0.00	1.00	0.00	40.00
sx6	1.00	0.50	1.00	0.00	0.00	1.00	45.00
Lower Bound	0.00	0.00	0.00				
Upper Bound	infinity	infinity	infinity				
Unrestr'd (y/n)?	n	n	n				

Iteration 2	Coef x1	Coef x2	Coef x2				
Basic	x1	x2	x3	sx4	sx5	sx6	Solution
z (max)	6.00	-7.00	0.00	15.00	0.00	0.00	630.00
x3	0.67	0.33	1.00	0.33	0.00	0.00	14.00
sx5	0.67	0.33	0.00	-0.67	1.00	0.00	12.00
sx6	0.33	0.17	0.00	-0.33	0.00	1.00	31.00
Lower Bound	0.00	0.00	0.00				
Upper Bound	infinity	infinity	infinity				

Figura 7: Ejercicio 5 en tora



Problem Title: Ejercicio 5

Nbr. of Variables: 3

No. of Constraints: 3

Editing Grid:
>>Click Maximize(Minimize)-cell to change it to Minimize(Maximize)
>>To DELETE, INSERT, COPY, or PASTE a column(row), click heading cell of target column(row), then invoke pull-down EditGrid menu
>>For INSERT mode, a single(double) click of target row/column will place new row/column after(before) target row/column.

INPUT GRID - LINEAR PROGRAMMING

	x1	x2	x3	Enter <, >, or =	R.H.S.
Var. Name	Coef x1	Coef x2	Coef x2		
Maximize	24.00	22.00	45.00		
Constr 1	2.00	1.00	3.00	<=	42.00
Constr 2	2.00	1.00	2.00	<=	40.00
Constr 3	1.00	0.50	1.00	<=	45.00
Lower Bound	0.00	0.00	0.00		
Upper Bound	infinity	infinity	infinity		
Unrestr'd (y/n)?	n	n	n		

Figura 8: Ejercicio 5 en tora